

Small Project 2 - Non-Autoregressive Architectures & Agentic Systems

Fundamentos de Aprendizagem Automática

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Small Project 2: Non-Autoregressive Architectures & Agentic Systems

1. Context & Objectives

While autoregressive generative AI (LLMs) dominates contemporary discussions, emerging paradigms address fundamental limits of autoregression (hallucinations, exposure bias, high latency, lack of grounded world dynamics): * **Joint Embedding Predictive Architectures (JEPA)**: Learning representations by predicting in latent feature space rather than input/pixel space (LeCun). * **World Models**: Learning compact latent transition dynamics for planning and control (Ha & Schmidhuber, Dreamer). * **Agentic AI Systems**: Autonomous goal-directed loops combining tool use, memory, and reflection.

Students will choose one of two tracks: * **Track A (Representation & Dynamics)**: Implement and evaluate a simplified JEPA or World Model architecture on a continuous control or visual prediction benchmark. * **Track B (Agentic AI Systems)**: Build a multi-step autonomous agent with tool execution, verification loops, and self-correction, evaluating failure modes and robustness against prompt perturbation.

2. Deliverables

1. **Source Code & Implementation**: Reproducible repository with environment configuration and automated tests.
2. **Technical Report (PDF, max 6 pages)**:

- Architectural formulation and design choices.
- Quantitative evaluation against baseline models.
- Ablation analysis (e.g. latent space dimension, tool failure recovery rate).

3. **Live Demonstration:** Brief 5-minute presentation in Class 14.

3. Important Dates & Submission

- **Release:** Week 9 (November 12-13, 2026)
- **Submission Deadline:** Class 14 (December 17-18, 2026)
- **Weight:** 25% of final course grade.